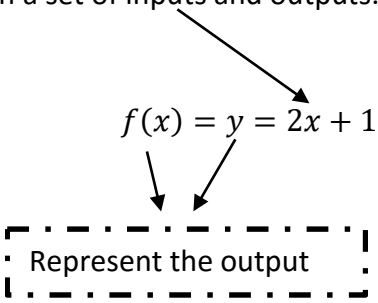
E-Classroom	**Name:**					
	Gr 10		**Date:**		**Time:**	$1\frac{1}{2}HR$
CAPS Reference	**Functions**					
Topic	**Linear Function**					

1. GOT IT! [15 MIN]

Function is a relation between a set of inputs and outputs.



We can get the desired output by implementing an input.

Example:

We read it as f of x is equal to $2x + 1$.

$$f(x) = 2x + 1$$

If the input is 1 the output will be 3. We calculate it by substituting 1 into the given equation.

$$f(1) = 2(1) + 1 = 2 + 1 = 3$$

$$\therefore (1; 3)$$

If it were asked to **reflect** the given equation across the $\boldsymbol{x - axis}$, remember to multiply the equation with -1.

$f(x) = 2x + 1$ $\quad reflection\ of\ x - axis\ \ -f(x) = 2x + 1$

$$f(x) = -2x - 1$$

If it were asked to **reflect** the given equation across the $\boldsymbol{y - axis}$, remember to multiply the $only\ the\ x$ with -1.

$f(x) = 2x + 1$ $\quad reflection\ of\ y - axis\ f(-x) = 2(-x) + 1$

$$f(x) = -2x + 1$$

Basic standard:

$$\boldsymbol{f(x) = mx + c}$$

$f(x)$: – represents the y – value

m: – represents the gradient

– Gradient formula $= \frac{y_2 - y_1}{x_2 - x_1}$

– indicates whether function will increase or decrease

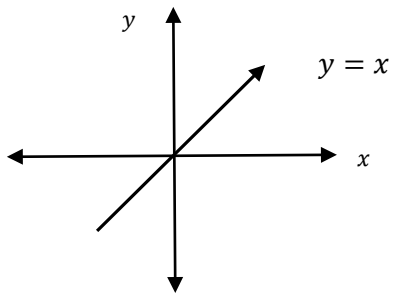
c: – represents the $y - intercept$

– indicates when the function will move up or down

Gradient of a Linear function.

Positive function; $m > 0$

Function is increasing

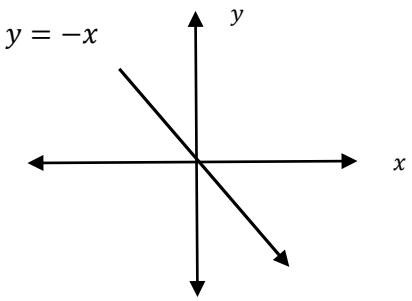


Domain: $x \in (-\infty; \infty)$

Range: $y \in (-\infty; \infty)$

Negative function; $m < 0$

Function is decreasing



Domain: $x \in (-\infty; \infty)$

Range: $y \in (-\infty; \infty)$

Drawing of a Linear function.

$$f(x) = 3x + 1$$

$\underline{y - intercept; x = 0}$

$y = 3(0) + 1$

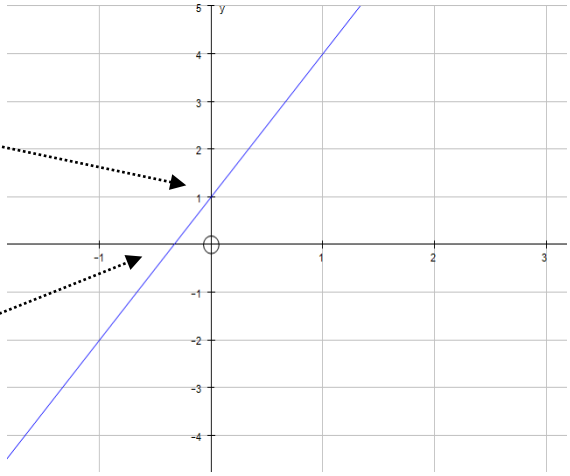
$= 1 \qquad (0; 1)$

$\underline{x - intercept; y = 0}$

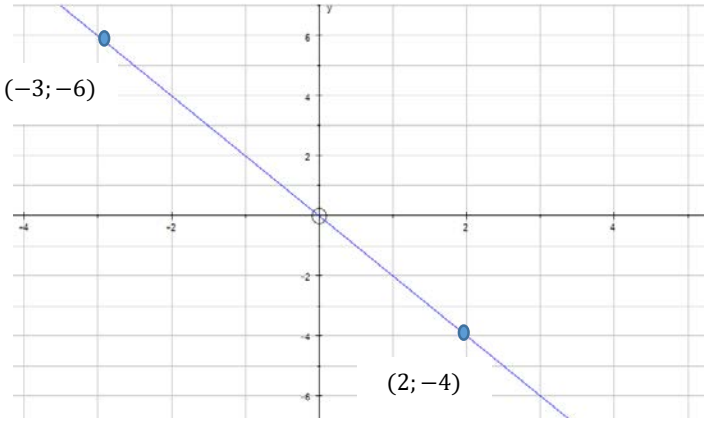
$0 = 3x + 1$

$-1 = 3x$

$\frac{-1}{3} = x \qquad \left(\frac{-1}{3}; 0\right)$



Calculating the function $f(x) = mx + c$



Equation $\therefore f(x) = -2x$

Step 1: Calculate the gradient

$x_1 \quad y_1 \qquad x_2 y_2$

$(-3; 6) \qquad (2; -4)$

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-4 - 6}{2 - (-3)}$$

$$= \frac{-10}{5} = -2$$

Step 2: Substitute $m = -2$

$$f(x) = -2x + c$$

Step3: Substitute $(2; -4)$ *in* $f(x) = -2x + c$

$-4 = -2(2) + c$

$-4 = -4 + c$

$c = -4 + 4$

$c = 0$

2. GO AHEAD [60 MIN]

2.1 Sketch the following on a set of axis and answer the questions that follows.

2.1.1 $f(x) = 4x + 2$

a) Write down the domain of f

b) Write down the gradient of f

c) Determine the value of x for which $f(x) < 0$

2.1.2 $g(x) = -2x + 1$

a) Write down the range of g

b) Write down the gradient of g

c) Determine the value of x for which $g(x) > 0$

2.1.3 $h(x) = -x - 1$

a) Write down the domain of h

b) Write down the gradient of h

c) Determine the value of x for which $h(x) > 0$

2.1.4 $k(x) = \frac{1}{3}x + 4$

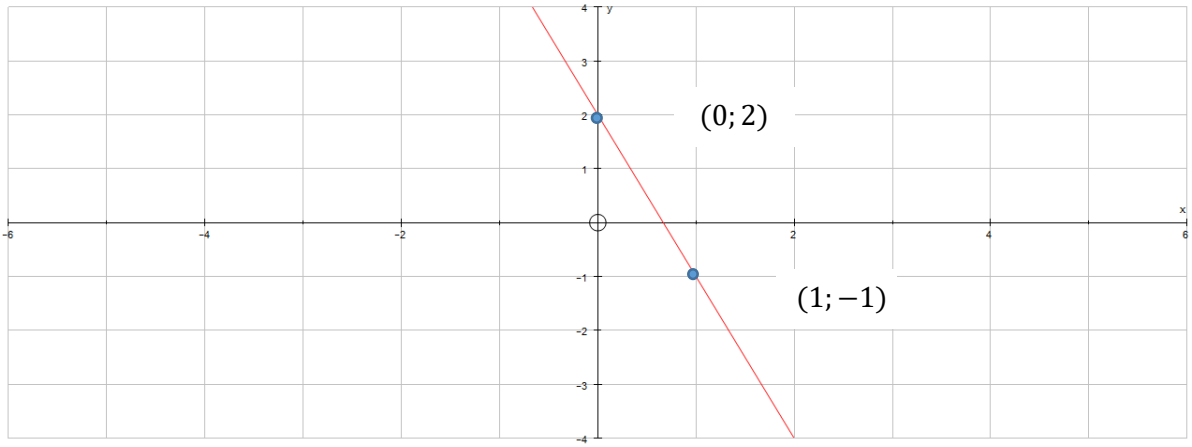
a) Write down the range of k

b) Write down the gradient of k

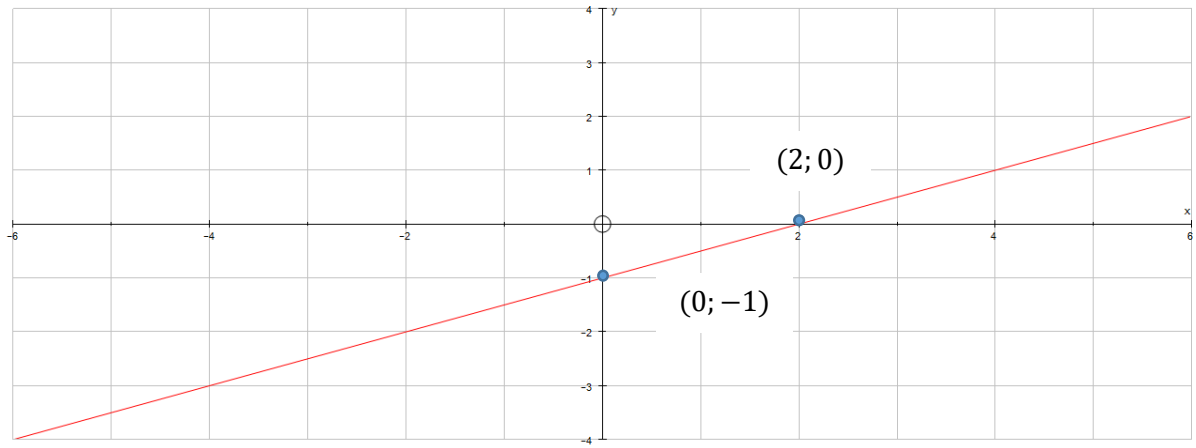
c) Determine the value of x for which $k(x) < 0$

2.2 Determine the equation of the following functions in the format $f(x) = mx + c$:

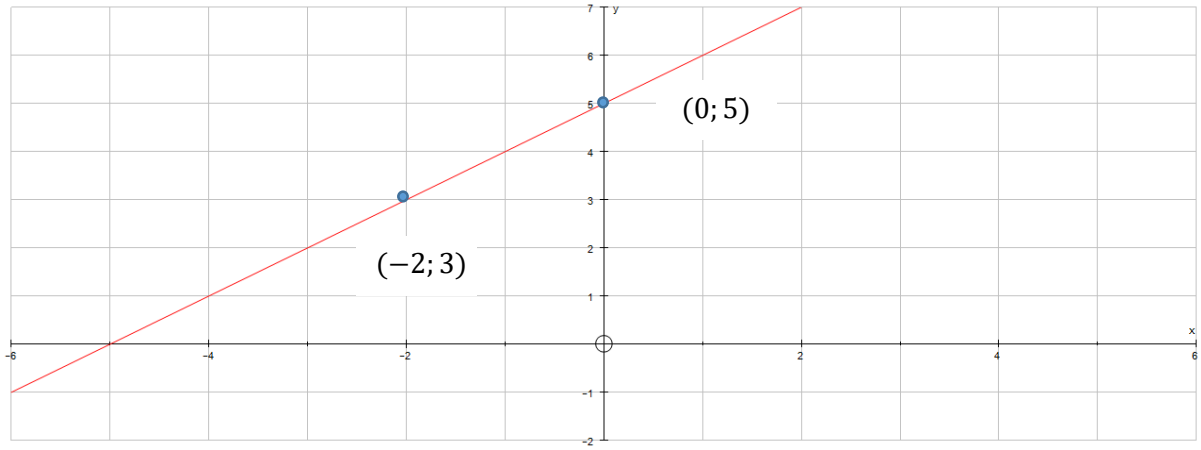
2.2.1



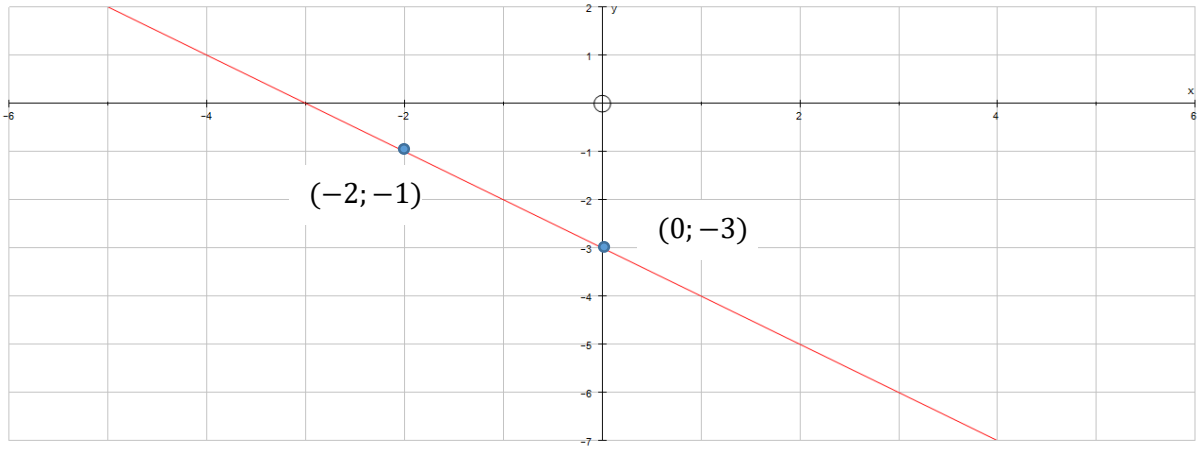
2.2.2



2.2.3



2.2.4



3. CHECK YOUR WORK![15 MIN]

4. HELPFUL VIDEO

https://www.youtube.com/watch?v=Y6zK_rPc290

https://www.youtube.com/watch?v=VXc7PjVl4UY